

WORKBOOK ASSIGNMENT-4

QUESTION1

Questions: Q1) Create a MySQL database named CollegeDB and create the following tables:

-- Create Database

```
CREATE DATABASE SeniorCollegeDB;
```

```
USE SeniorCollegeDB;
```

-- Create Student Table

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50),  
    City VARCHAR(30)  
);
```

-- Create Course Table

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(50),  
    Fees DECIMAL(10,2)  
);
```

-- Create Enrollment Table

```
CREATE TABLE Enrollment (  
    EnrollmentID INT PRIMARY KEY,  
    StudentID INT,  
    CourseID INT,
```

EnrollmentDate DATE,
 FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
 FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);



QUESTION2

Q2) Insert at least 6 records into the Student table, 4 records into the Course table, and 5 records into the Enrollment table. Make sure:

Requirement
At least one student should not have enrollment
At least one course should not have any enrolled student

-- Insert into Student table

INSERT INTO Student VALUES

- (1, 'Alice', 'NY'),
- (2, 'Bob', 'LA'),
- (3, 'Charlie', 'SF'),
- (4, 'Dave', 'CHI'),
- (5, 'Eve', 'MIA'),
- (6, 'Frank', 'SEA');

-- Insert into Course table

INSERT INTO Course VALUES

```
(101, 'Math', 500.00),
(102, 'Science', 600.00),
(103, 'Art', 300.00),
(104, 'History', 400.00);
```

-- Insert into Enrollment table

INSERT INTO Enrollment VALUES

```
(1001, 1, 101, '2023-01-01'),
(1002, 2, 101, '2023-01-02'),
(1003, 3, 102, '2023-01-03'),
(1004, 4, 103, '2023-01-04'),
(1005, 5, 103, '2023-01-05');
```



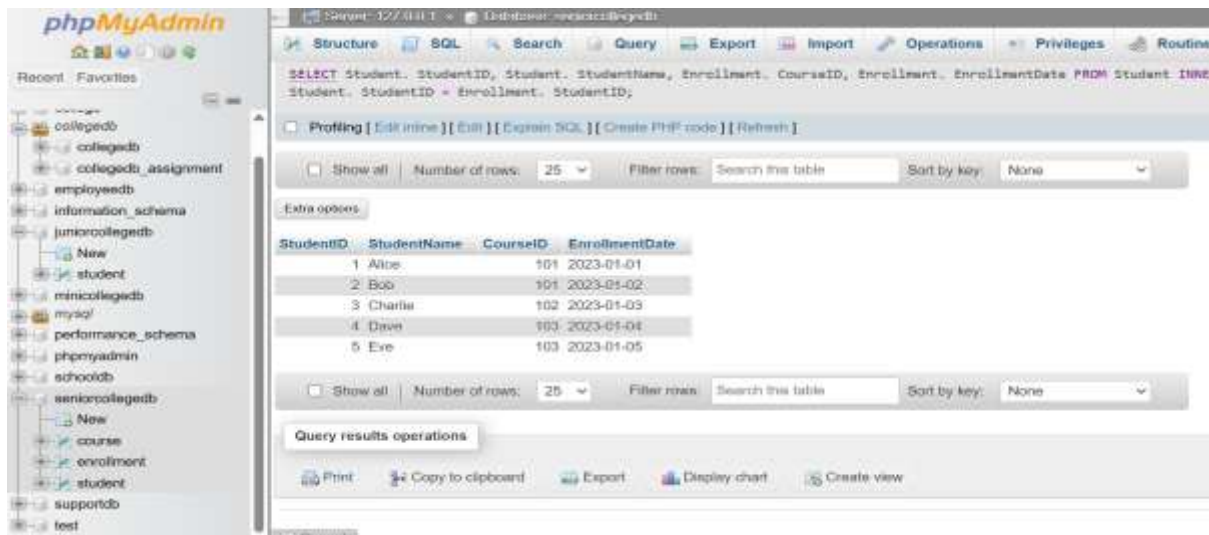
QUESTION3

Q3) Write a MySQL query using INNER JOIN to display students who have enrolled in courses. The output should include:

Column
StudentID
StudentName
CourseID
EnrollmentDate

```
SELECT Student.StudentID,
Student.StudentName,
```

Enrollment. CourseID,
 Enrollment. EnrollmentDate
 FROM Student
 INNER JOIN Enrollment ON
 Student. StudentID =
 Enrollment. StudentID;



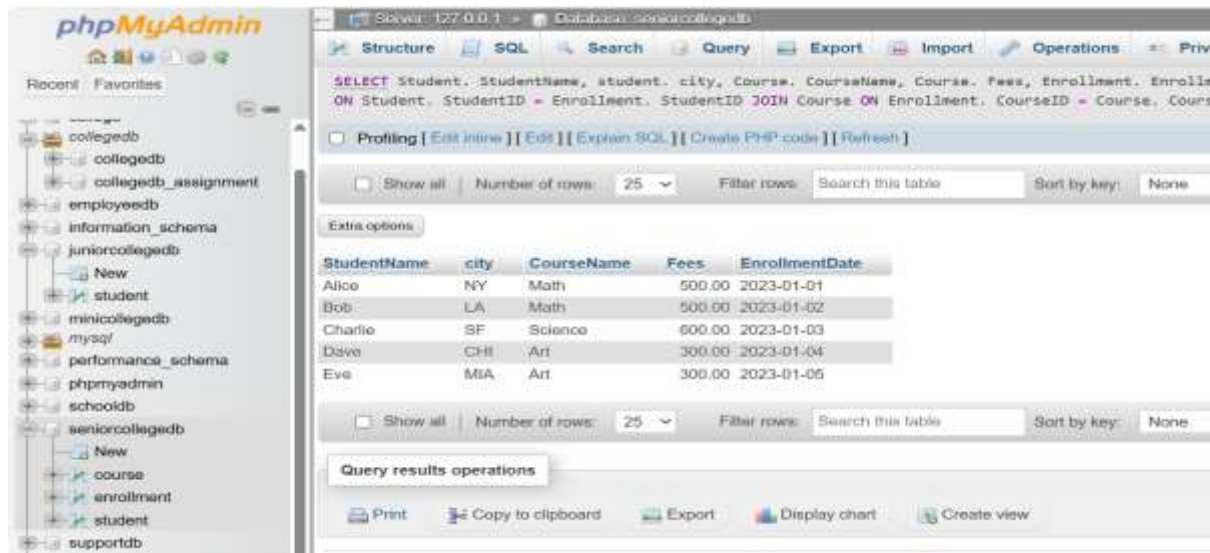
QUESTION4

Q4) Write a MySQL query to display student name, city, course name, fees, and enrollment date using multiple tables. Use:

Tables Required
Student
Course
Enrollment

SELECT Student. StudentName,
 student. city,
 Course. CourseName,
 Course. Fees,
 Enrollment. EnrollmentDate
 FROM Student
 JOIN Enrollment ON
 Student. StudentID =

Enrollment. StudentID
 JOIN Course ON Enrollment. CourseID
 = Course. CourseID;

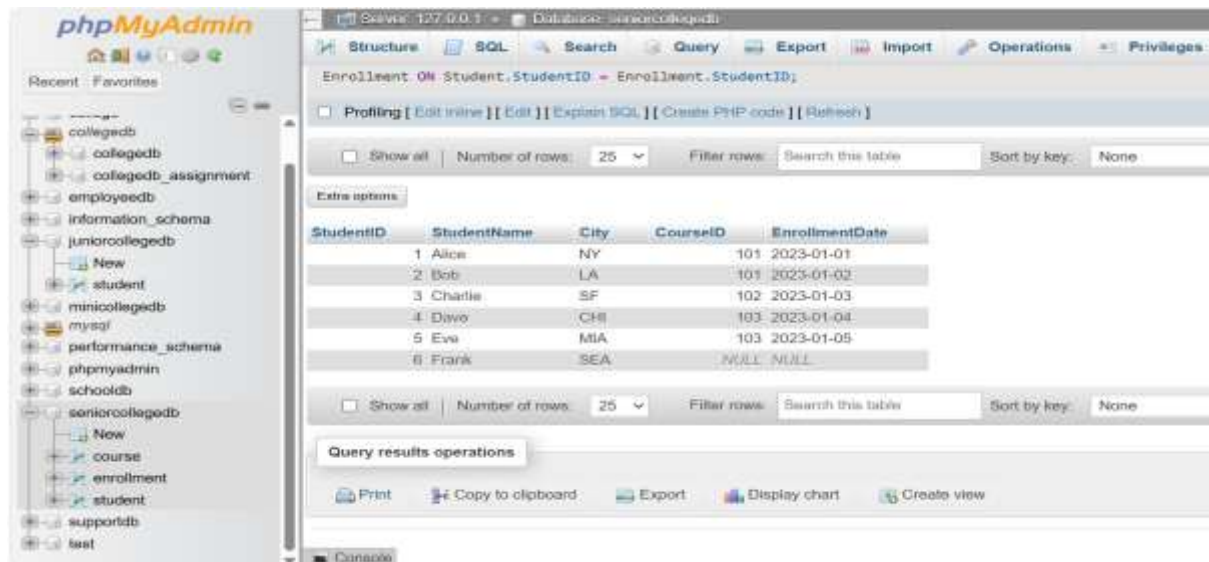


QUESTIONS

Q5) Write a MySQL query using LEFT JOIN to display all students, including those who have not enrolled in any course. The output should include:

Column
StudentID
StudentName
City
CourseID
EnrollmentDate

SELECT Student. StudentID,
 Student. StudentName,
 Student. City,
 Enrollment. CourseID,
 Enrollment. EnrollmentDate
 FROM
 Student
 LEFT JOIN Enrollment
 ON Student.StudentID = Enrollment.StudentID;



QUESTION6

Q6) Write a MySQL query using RIGHT JOIN to display all courses, including courses that have no enrolled students. The output should include:

Column
StudentName
CourseID
CourseName
Fees

```

SELECT Student.StudentName,
Course. CourseID,
Course. CourseName,
Course. Fees
FROM Student
JOIN Enrollment ON
Student.StudentID = Enrollment.StudentID
RIGHT JOIN Course ON
Enrollment. CourseID = Course. CourseID;

```

phpMyAdmin

Server: 127.0.0.1 - Database: seniorcollegedb

Structure SQL Search Query Export Import Operations Privileges Roles

Enrollment.StudentID RIGHT JOIN Course ON Enrollment.CourseID = Course.CourseID;

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentName	CourseID	CourseName	Fees
Alice	101	Math	500.00
Bob	101	Math	500.00
Charlie	102	Science	600.00
Dave	103	Art	300.00
Eve	103	Art	300.00
NULL	104	History	400.00

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)